

■ Smart Recruitment Assistant

Generative AI-powered Resume Screening and Shortlisting Automation

■ Problem Statement

Recruiters spend excessive time screening resumes and matching candidates with job descriptions. Manual shortlisting is often slow, inconsistent, and prone to human bias.

■ Objective

To build an AI-driven recruitment assistant that:

- Analyzes job descriptions and resumes for best-fit matching.
- Provides top N candidate recommendations automatically.
- Automates personalized shortlisting emails to candidates.

■■ Workflow Overview

The automation pipeline (as shown in the attached visual) is designed to streamline the recruitment process using Google Gemini AI, form automation, and email integration.

■ Pipeline Components

■ *On Form Submission*

Triggered when a recruiter submits a form containing a job description and uploaded candidate resumes (PDF format). Captures and passes input data to the next stage.

■ *Code in JavaScript*

Processes form inputs, formats data, and prepares it for AI analysis. Extracts metadata like candidate name, email, and skills.

■ *Extract from File (Extract from PDF)*

Extracts text content from uploaded resume PDFs. Sends clean resume text for semantic analysis.

■ *AI Agent (Google Gemini Chat Model + Simple Memory)*

Uses Google Gemini AI for natural language understanding. Compares resume content with job descriptions. Scores and ranks candidates based on skill and experience match.

■ **Code in JavaScript (Post-Processing)**

Sorts and filters AI output. Prepares the top N candidate recommendations in structured format.

■ **AI Agent 2 (Final Summary)**

Generates a personalized message or summary for each shortlisted candidate. Uses generative text to make communication more human-like.

■ **Send a Message (Gmail Integration)**

Sends automated, customized email notifications to shortlisted candidates. Example: "Dear [Name], based on your profile and experience, you've been shortlisted for the [Role Name] position at [Company Name]. We'll contact you soon for the next steps."

■ **Tech Stack**

- Google Gemini AI – Resume & JD analysis, text generation
- JavaScript (Node.js) – Data formatting, logic processing
- Form Triggers – Input automation
- PDF Extractor – Resume content extraction
- Gmail API – Automated candidate communication
- AI Agent Memory – Contextual conversation and learning

■ **Key Features**

- Automated resume parsing and analysis
- Intelligent JD-to-resume matching using AI
- Ranked candidate recommendations
- Personalized shortlisting emails
- Fully automated recruitment pipeline

■ **Future Enhancements**

- Integration with LinkedIn or job portals for live candidate sourcing
- Enhanced semantic search and ranking
- Candidate dashboard for real-time status updates
- Bias detection and diversity scoring

■■■■ **Developed By**

Project: Smart Recruitment Assistant

DeveloPED BY TECH TITANS

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